

ALCONBURY, United Kingdom

WMO: 035620

Lat: 52.367N

Lon: 0.217W

Elev: 157

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 83-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.5	26.6	15.8	12.3	25.1	19.2	14.5	28.9	30.4	46.0	27.3	45.9	5.6	70	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	80.2	65.6	75.6	63.4	73.3	61.9	67.1	77.1	65.0	73.9	63.2	70.9	8.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	86.3	70.6	61.1	80.9	68.9	59.4	76.0	67.0	31.5	76.6	30.0	74.2	28.6	71.4	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	20.7	18.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	17.7	68.9	5.4	2.2	13.8	70.5	10.7	71.8	7.6	73.1	3.7	74.7	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	39.6	38.7	43.6	46.9	52.8	57.6	62.7	62.4	57.4	51.6	43.9	42.1
	DBStd	9.95	6.29	6.60	5.18	5.30	4.97	4.95	4.88	5.11	4.78	5.84	5.58	6.32
	HDD50	1501	323	319	207	124	25	2	0	0	3	51	193	254
	HDD65	5545	787	737	664	543	379	227	105	113	230	418	633	709
	CDD50	1511	2	2	8	31	112	230	395	386	226	99	11	9
	CDD65	80	0	0	0	0	1	5	35	34	3	2	0	0
	CDH74	565	0	0	0	0	13	47	248	232	19	6	0	0
	CDH80	95	0	0	0	0	0	2	44	47	1	1	0	0
Wind	WSAvg	8.2	9.9	9.4	9.5	8.8	8.1	7.3	7.0	6.9	7.6	7.9	8.0	8.5
Precipitation	PrecAvg	21.1	1.7	1.2	1.6	1.7	1.8	1.9	1.8	2.0	2.0	1.8	1.9	1.8
	PrecMax	28.7	3.5	2.9	3.2	3.7	4.2	4.8	4.8	4.2	4.8	5.2	3.9	3.9
	PrecMin	11.0	0.3	0.4	0.1	0.3	0.2	0.1	0.3	0.4	0.2	0.1	0.9	0.4
	PrecStd	3.5	0.7	0.6	0.9	0.9	1.0	1.1	1.0	1.0	1.3	1.3	0.8	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.1	62.2	68.1	75.4	79.0	84.5	85.7	76.8	71.6	60.8	57.4
		MCWB	52.4	53.1	53.7	57.7	61.5	64.6	68.3	67.3	62.3	62.3	56.5	52.8
	2%	DB	53.5	53.5	57.0	63.0	70.1	74.8	80.4	80.2	71.9	65.9	56.8	55.5
		MCWB	50.3	49.7	51.4	55.2	59.9	63.1	66.0	65.2	61.0	59.1	53.1	51.5
	5%	DB	51.4	51.4	53.9	59.1	66.1	71.3	76.6	75.5	69.5	62.7	54.0	53.6
		MCWB	48.2	47.9	49.4	52.9	57.0	60.7	64.4	63.2	60.1	57.2	50.6	49.9
	10%	DB	48.4	48.1	51.8	55.6	62.6	67.8	73.2	73.0	66.0	60.7	52.2	51.7
		MCWB	46.1	45.8	47.6	49.9	54.7	59.0	62.4	61.7	58.2	55.5	48.8	48.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.0	55.0	59.6	63.8	66.5	71.7	68.1	65.0	63.1	56.2	54.6
		MCDB	55.1	56.2	59.7	66.0	72.9	75.8	81.0	81.4	71.5	70.2	59.9	56.8
	2%	WB	50.9	50.3	52.3	56.2	60.6	63.7	68.1	66.3	62.4	59.7	54.0	52.3
		MCDB	53.3	52.8	56.4	61.4	68.6	71.9	76.8	76.8	69.2	64.5	56.8	55.0
	5%	WB	48.4	48.1	50.4	53.5	58.0	61.4	65.6	64.5	60.8	57.5	51.9	50.4
		MCDB	50.6	50.3	53.3	58.8	65.1	69.3	74.1	73.9	67.4	61.9	54.4	53.6
	10%	WB	46.1	45.6	48.5	50.9	55.5	59.5	63.3	62.8	59.3	55.6	49.9	48.4
		MCDB	48.2	47.6	51.5	55.6	61.0	67.0	70.6	70.7	64.7	59.7	52.5	51.6
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.5	11.8	13.5	14.8	14.8	16.2	16.3	14.0	11.9	9.6	7.8
		MCDBR	11.0	12.0	14.3	17.6	21.5	20.6	21.9	21.0	18.6	13.6	10.8	8.8
		MCWBR	10.3	10.0	10.8	11.7	12.8	12.0	11.3	10.2	10.8	9.3	9.4	8.1
	5% WB	MCDBR	11.1	11.6	13.6	16.8	20.0	18.7	20.1	19.3	16.2	12.4	10.8	9.1
		MCWBR	11.0	10.3	11.0	11.5	12.8	11.6	11.5	10.1	10.7	9.3	9.9	8.7
Clear-Sky Solar Irradiance	taub	0.323	0.340	0.373	0.398	0.402	0.405	0.404	0.393	0.372	0.354	0.327	0.313	
	taud	2.369	2.386	2.284	2.227	2.254	2.268	2.282	2.336	2.397	2.435	2.416	2.350	
	Ebn at Noon	212	245	257	265	269	269	267	264	256	236	209	193	
	Edh at Noon	19	25	33	40	41	41	39	35	30	23	18	16	
All-Sky Solar Radiation	RadAvg	240	458	801	1237	1517	1626	1559	1294	971	565	295	194	
	RadStd	22	59	95	134	102	151	166	94	85	49	32	21	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page